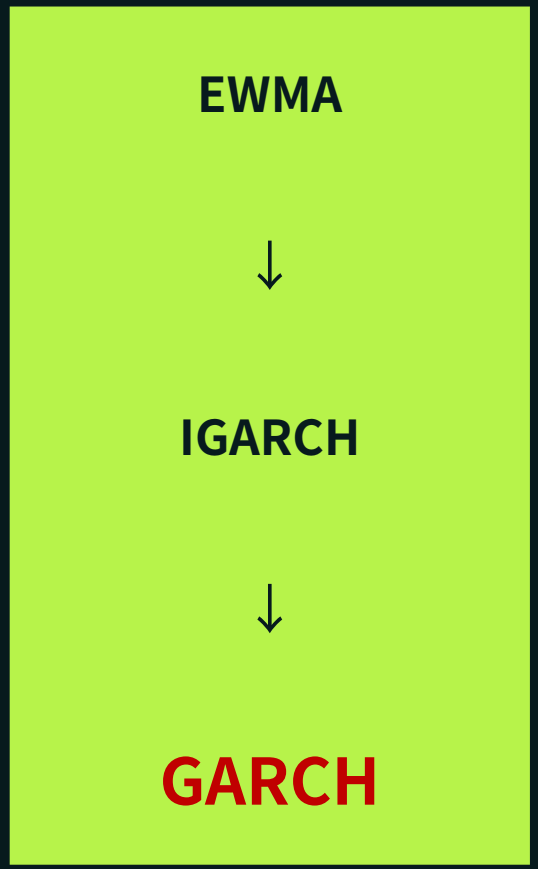
GARCH vs. EWMA Anchored or Not?

GARCH vs EWMA — is being tied to a long-run level a constraint or a safeguard?



Advanced Financial Time Series

Both models set today's variance as a weighted average of yesterday's information

Basic idea — conditional variance: the variance of today's return given information up to yesterday

Ingredient ①

Yesterday's squared return

- A big move yesterday means a big move today
- The share of news

Ingredient ②

Yesterday's variance estimate

- Volatility built up through yesterday
- The share of memory

Ingredient ③ — GARCH only

Long-run average variance

- The level it returns to in the long run
- The share of the anchor

One difference — GARCH has an anchor, the long-run level; EWMA does not

GARCH updates variance with three parameters, EWMA with a single λ

GARCH(1,1) is Bollerslev (1986); EWMA is J.P. Morgan RiskMetrics (1994) — λ = 0.94 for daily data

The two update rules

$$\text{GARCH}(1,1): \quad \sigma_t^2 = \omega + \alpha\, r_{t-1}^2 + \beta\, \sigma_{t-1}^2, \qquad \bar{\sigma}^2 = \frac{\omega}{1-\alpha-\beta}$$

$$\text{EWMA}: \quad \sigma_t^2 = (1-\lambda)\, r_{t-1}^2 + \lambda\, \sigma_{t-1}^2, \qquad \lambda = 0.94$$

How to read them

- ω: constant (anchor size)
- α: reaction to news
- β: memory persistence
- $\bar{\sigma}^2$: long-run variance
- λ: decay, fixed (not estimated)

GARCH is estimated from the data; EWMA fixes λ and is used right away

Written as weights, EWMA is GARCH with the anchor's share set to 0

Rewriting ω in the GARCH equation as $(1-\alpha-\beta)\cdot\bar{\sigma}^2$ gives a weighted average of three shares

Weighted average of three shares

$$\text{GARCH}: \quad \sigma_t^2 = \underbrace{(1-\alpha-\beta)}_{\text{anchor}}\,\bar{\sigma}^2 + \underbrace{\alpha}_{\text{news}}\,r_{t-1}^2 + \underbrace{\beta}_{\text{memory}}\,\sigma_{t-1}^2$$

$$\text{EWMA}: \quad \sigma_t^2 = \underbrace{0}_{\text{no anchor}}\cdot\bar{\sigma}^2 + (1-\lambda)\,r_{t-1}^2 + \lambda\,\sigma_{t-1}^2$$

How to read them

- anchor: long-run level
- news: yesterday's squared return
- memory: yesterday's variance
- The three weights sum to 1

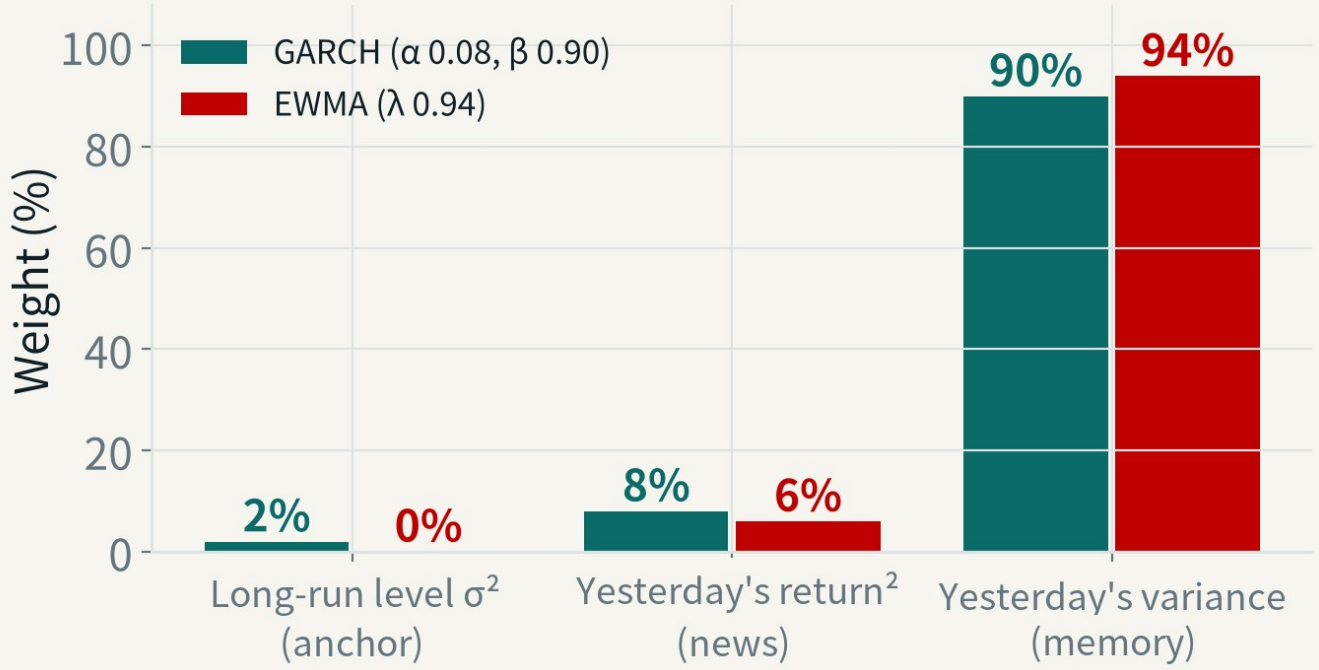
EWMA sets the anchor's weight to 0 and splits that share between news and memory

The anchor's share is only 2%, but it builds up every day and pulls the forecast

GARCH α 0.08, β 0.90 / EWMA λ 0.94 — comparing the weights

How to read them

- GARCH: puts 2% on the long-run level
- EWMA: gives everything to recent information
- The news and memory shares are almost the same



One-day updates are similar, but the anchor's 2% makes a difference as the horizon grows

EWMA is a special case of GARCH with two constraints

IGARCH (Integrated GARCH) — a model in which shocks never fade and persist forever

$$\text{EWMA} = \text{GARCH}(1,1) \text{ with } \omega = 0, \quad \alpha = 1 - \lambda, \quad \beta = \lambda \quad \Rightarrow \quad \alpha + \beta = 1 \quad (\text{IGARCH})$$

	GARCH(1,1)	EWMA
Parameters to estimate	**ω, α, β (three)**	**One λ, usually fixed at 0.94**
α + β	**Less than 1 (stationary)**	**Exactly 1 (non-stationary)**
Long-run average variance	**Exists ($\bar{\sigma}^2$)**	**Not defined**
Effect of a shock	**Fades gradually**	**Persists forever**

'Free' means it has no anchor, not that the model is more general

The difference shows in forecasts — GARCH reverts, EWMA stays put

h-step-ahead variance forecast — if α + β < 1, (α + β) to the power h shrinks to 0

h-step-ahead forecast

$$\text{GARCH}: \quad E_t[\sigma^2_{t+h}] = \bar{\sigma}^2 + (\alpha+\beta)^{h-1}\left(\sigma^2_{t+1} - \bar{\sigma}^2\right) \longrightarrow \bar{\sigma}^2$$

$$\text{EWMA}: \quad E_t[\sigma^2_{t+h}] = \sigma^2_{t+1} \quad (\text{flat for all } h)$$

How to read them

- GARCH: the gap to the long-run level shrinks by (α+β) each period
- EWMA: keeps today's estimate for every horizon

Short-term forecasts are almost the same; the longer the horizon, the further the two models diverge

Simple example: one-day-ahead variance is nearly the same in both

Yesterday's volatility 2%, yesterday's return 3% (a shock), long-run volatility 1% — units are %²

One-day-ahead variance

$$\textbf{Inputs} \quad \bar{\sigma}^2 = 1,\ \alpha = 0.08,\ \beta = 0.90 \ \Rightarrow\ \omega = 0.02; \quad \lambda = 0.94$$
$$\sigma^2_{t-1} = 4\ (2\%), \quad r_{t-1} = 3\% \ \Rightarrow\ r^2_{t-1} = 9$$
$$\text{GARCH}: \ \sigma^2_t = 0.02 + 0.08 \times 9 + 0.90 \times 4 = 4.34 \quad (\sigma_t \approx 2.08\%)$$
$$\text{EWMA}: \ \sigma^2_t = 0.06 \times 9 + 0.94 \times 4 = 4.30 \quad (\sigma_t \approx 2.07\%)$$

How to read them

- GARCH 4.34, EWMA 4.30
- In volatility, 2.08% vs 2.07%
- Because the news and memory weights are similar

For a one-day-ahead VaR, either model gives similar results

Stretch the same example to 20 and 60 days ahead, and only GARCH comes down to 1%

Same numbers as before — α + β = 0.98, so the gap of 3.34 (= 4.34 − 1) shrinks by 0.98 each day

Multi-period variance forecast

$$\text{GARCH},\ h=20:\ 1+0.98^{19}\times(4.34-1)\approx 1+0.68\times 3.34=3.27\quad(1.81\%)$$
$$\text{GARCH},\ h=60:\ 1+0.98^{59}\times 3.34\approx 1+0.30\times 3.34=2.01\quad(1.42\%)$$
$$\text{EWMA},\quad \text{any } h:\ 4.30\quad(2.07\%)$$

How to read them

- 20 days: 1.81 vs 2.07
- 60 days: 1.42 vs 2.07
- Half-life about 34 days

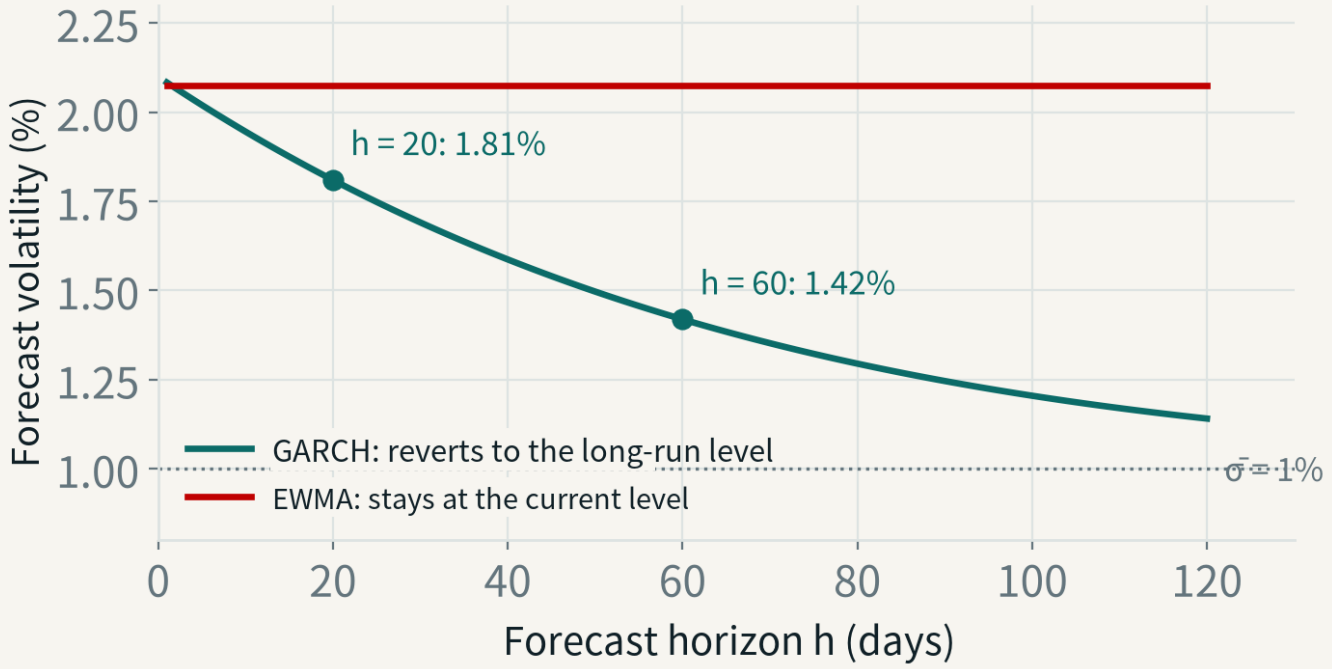
As the horizon grows, 'anchored or not' decides the result

Plotted against the horizon, GARCH comes down while EWMA stays flat

Term structure of forecast volatility — numbers from the previous example

How to read them

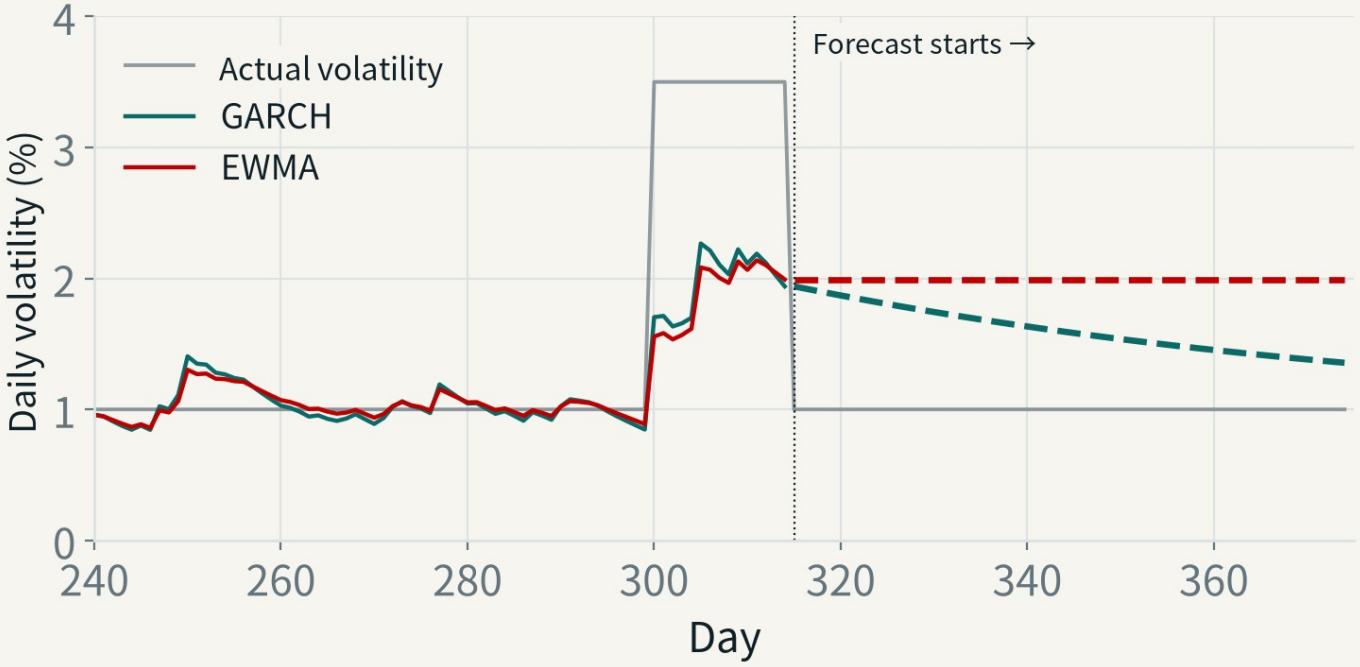
- Green: converges to 1%
- Red: flat at 2.07%
- The gap widens for option maturities and monthly VaR



Same in the short run, different in the long run

After a temporary shock, GARCH — which returns to the long-run level — is right

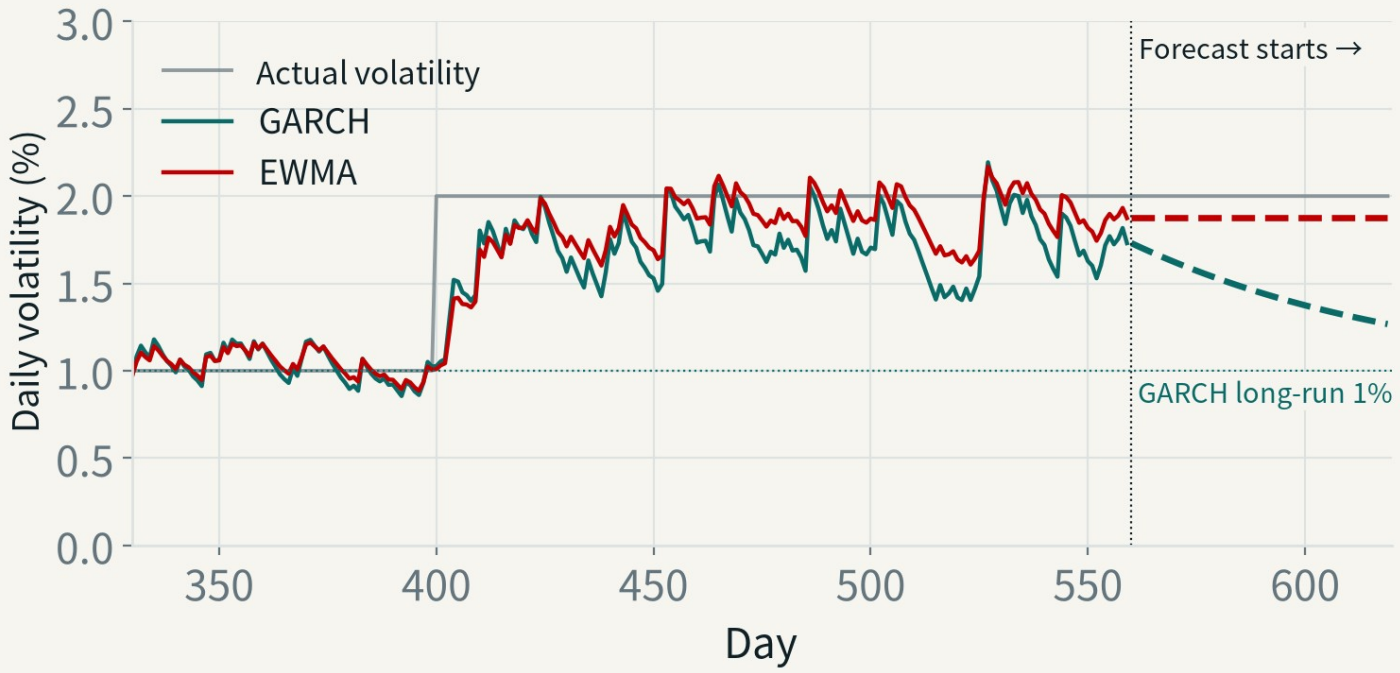
Simulation — 1% normally, a 15-day spike to 3.5%, then back to 1% / solid: estimate, dashed: 60-day-ahead forecast



EWMA keeps dragging past shocks along and over-forecasts volatility

If the long-run level itself changes, EWMA — not tied to an anchor — is right

Simulation — on day 400 volatility rises permanently from 1% to 2%; GARCH's ω stays fixed at the old 1% level



GARCH is pulled down to the old long-run level of 1% and under-forecasts volatility

GARCH wins when shocks are temporary; EWMA wins when the regime changes

Choosing in practice — look at the forecast horizon and the stability of the long-run level

Situation	Better model	Why
VaR one to a few days ahead	**Either (EWMA is simpler)**	**Short-term forecasts are almost the same**
Option maturities, monthly risk	**GARCH**	**Reflects mean reversion**
Regime shifts, structural change	**EWMA**	**Not tied to an old long-run level**
Many assets at once	**EWMA**	**Uses the same λ without estimation**
Estimated α + β near 0.99	**Either**	**The two models become effectively the same**

If GARCH's anchor is a worry, re-estimate ω on a rolling window

EWMA = GARCH without the anchor

Same in the short run, different in the long run

Temporary shocks: GARCH; regime shifts: EWMA

Freedom has a price
No anchor, no place to return to